

Sample Surveys

Mid-Semester Question Bank, with Full Solutions

B.Stat. 3rd year, Semester V — Classes 1–9

This course has **no previous-year papers, assignments or homeworks**, so the questions below are constructed directly from the lecture slides. Mark weights follow the expected pattern: mostly 8–12, with shorter 5-mark items on definitions and recall.

Section	Topic		Qs	Marks
2	Concepts and vocabulary	<i>(Class 1)</i>	6	39
3	Designs, selection and inclusion probabilities	<i>(Class 2)</i>	5	41
4	Estimation under SRSWR and SRSWOR	<i>(Classes 2–3)</i>	7	64
5	Effective sample size and proportions	<i>(Class 3)</i>	4	36
6	Accuracy, confidence intervals, sample size	<i>(Classes 3–5)</i>	11	103
7	Stratified random sampling	<i>(Classes 6–8)</i>	8	86
8	Systematic sampling	<i>(Classes 8–9)</i>	8	80
9	Applied and mixed problems		4	44
Total			53	493

Solutions begin at §10 and mirror the question numbering, shifted by 8: question §7.3 is solved in §15.3, and so on. **Attempt first.**

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1 Syllabus map: what each class covered

Class	Content
1	Motivation and vocabulary. Mahalanobis, jute production, the NSS (1950), IPNS, pilot surveys; the Mahabharata anecdote. The ABCD : observation unit, target population, sampled population, sampling unit, sampling frame. Census vs. sample survey. The Literary Digest 1936 failure . What sampling buys. Total error = sampling + non-sampling . Probability vs. non-probability sampling; convenience, judgment, self-selection, snowball, quota.
2	Finite-population framework: $U = \{1, \dots, N\}$, Y_j fixed , parameters $\bar{Y}$, σ^2 . Sampling design $p: \mathcal{S} \rightarrow [0, 1]$; sampling scheme . SRSWR and SRSWOR. Implementation: lottery, random number tables (rejection and remainder methods), <code>sample()</code> . SRSWR design $p(s) = 1/N^n$, $\pi_j = 1 - (1 - 1/N)^n$. SRSWR estimation: $\bar{y}$ unbiased, $\text{Var}(\bar{y}) = \sigma^2/n$, $\mathbb{E}(s^2) = \sigma^2$. SRSWOR design $p(s) = 1/\binom{N}{n}$, $\pi_j = n/N$, $\pi_{jj'} = \frac{n(n-1)}{N(N-1)}$.
3	SRSWOR estimation: $\text{Var}(\bar{y}) = \frac{N-n}{N-1} \frac{\sigma^2}{n} = (1-f) \frac{S^2}{n}$, finite population correction ; $\mathbb{E}(s^2) = S^2$. SRSWOR beats SRSWR for $n \geq 2$; when SRSWR is used (bootstrap). Effective sample size n_{eff} , $\bar{y}_{\text{eff}}$. Total, variance, covariance. Proportions . Accuracy: SE, CV/RSD , Statistics Canada guidelines.
4	Confidence intervals : why exact CIs are impossible; Hájek (1960) ; the approximate CI; t_{n-1} for small n . Choosing n : margin of error, absolute e and relative r ; $n_{WR} = z_{\alpha/2}^2 \sigma^2 / e^2$, $n_{WOR} \approx n_{WR} / (1 + n_{WR}/N)$; proportions and the $P(1-P) \leq \frac{1}{4}$ bound.
5	Worst case $P = 0.5$ ($e = 0.05 \Rightarrow n = 385$). Supplying S^2 : worst-case, pilot survey , previous studies, guess. Relative error $\Rightarrow$ the CV . Testing perspective : α , β , power, effect size Δ ; $n_{WR} = (z_\alpha + z_\beta)^2 \sigma^2 / \Delta^2$; proportions. Worked: diabetes ($n = 31$), Framingham LDL ($n = 809$).
6	Stratified random sampling . Why SRS can fail (representative only <i>in expectation</i>); auxiliary information. Setup: H strata, N_h , n_h , $\sum N_h = N$, $\sum n_h = n$. $\bar{y}_{st} = \frac{1}{N} \sum N_h \bar{y}_h$ is unbiased; $\text{Var}(\bar{y}_{st}) = \frac{1}{N^2} \sum N_h^2 (1-f_h) \frac{S_h^2}{n_h}$; variance estimation and CI. Allocation : equal, proportional.
7	Comparison of variances : the exact identity for $\text{Var}_{SRS}(\bar{y}) - \text{Var}_{st}(\bar{y}_{st})$ under proportional allocation; when stratification helps (small WSS, large BSS). Neyman allocation $n_h \propto N_h S_h$, and that it minimises $\text{Var}(\bar{y}_{st})$ subject to $\sum n_h = n$. Optimal allocation with a cost constraint : $n_h \propto N_h S_h / \sqrt{C_h}$.
8	Remarks on stratification (strata-level parameters, different frames and field procedures per stratum, formation of strata). Systematic sampling : $N = nk$, random start in $\{1, \dots, k\}$, sampling interval k ; uses; behaviour under sorted vs. periodic frames. Estimation: $\bar{y}$ unbiased, $\text{Var}(\bar{y}) = \sigma_b^2$ (between-cluster). Intraclass correlation ρ_c and $\text{Var}(\bar{y}) = \frac{\sigma^2}{n} \{1 + (n-1)\rho_c\}$.

Class	Content
9	Systematic sampling inference. The variance cannot be estimated unbiasedly from one sample (a cluster-level sample of size 1). Pretending it is SR-SWOR: $\hat{V} = (1-f)s^2/n$ is biased , unbiased iff $\rho_c = -\frac{1}{N-1}$. Alternatives: m independent systematic samples (IPNS), multiple random starts, splitting one sample, Wolter (1984) successive-difference estimator. N not a multiple of n; circular systematic sampling .

The results that carry the course

$$\text{SRSWR: } \mathbb{E}(\bar{y}) = \bar{Y}, \quad \text{Var}(\bar{y}) = \frac{\sigma^2}{n}, \quad \mathbb{E}(s^2) = \sigma^2.$$

$$\text{SRSWOR: } \text{Var}(\bar{y}) = \frac{N-n}{N-1} \cdot \frac{\sigma^2}{n} = (1-f)\frac{S^2}{n}, \quad \mathbb{E}(s^2) = S^2, \quad S^2 = \frac{N}{N-1}\sigma^2.$$

$$\text{Stratified: } \bar{y}_{st} = \sum_h W_h \bar{y}_h, \quad \text{Var}(\bar{y}_{st}) = \frac{1}{N^2} \sum_h N_h^2 (1-f_h) \frac{S_h^2}{n_h}, \quad W_h = \frac{N_h}{N}.$$

$$\text{Allocation: } \text{prop. } n_h = nW_h; \quad \text{Neyman } n_h = \frac{nN_h S_h}{\sum N_{h'} S_{h'}}; \quad \text{cost } n_h \propto \frac{N_h S_h}{\sqrt{C_h}}.$$

$$\text{Systematic: } \text{Var}(\bar{y}) = \sigma_b^2 = \frac{\sigma^2}{n} \{1 + (n-1)\rho_c\}, \quad -\frac{1}{n-1} \leq \rho_c \leq 1.$$

$$\text{Sample size: } n_{WR} = \frac{z_{\alpha/2}^2 \sigma^2}{e^2} \text{ or } \frac{(z_\alpha + z_\beta)^2 \sigma^2}{\Delta^2}, \quad n_{WOR} \approx \frac{n_{WR}}{1 + n_{WR}/N}.$$

Trap — the three “opposite” facts that get mixed up

- **Stratified** sampling wants strata *internally homogeneous* (small S_h) and strata means far apart. **Systematic** sampling wants clusters *internally heterogeneous* (small ρ_c) and clusters resembling one another. **Exactly opposite.**
- s^2 is unbiased for σ^2 under SRSWR but for S^2 under SRSWOR. Same statistic, different target.
- Under systematic sampling $\bar{y}$ is unbiased, but its *variance* cannot be estimated unbiasedly at all — a distinction worth stating explicitly.

2 Concepts and vocabulary

2.1

[5 marks]

Define, and illustrate each with the running example “*estimate the average monthly out-of-pocket medical expenditure of all adults in West Bengal*”: (i) observation unit, (ii) target population, (iii) sampled population, (iv) sampling unit, (v) sampling frame. Explain what goes wrong when the sampled population differs from the target population.

2.2

[5 marks]

Distinguish a *census* from a *sample survey*. Give three distinct reasons why a sample survey can be **more accurate** than a complete enumeration.

2.3 [8 marks]

In 1936 the *Literary Digest* mailed 10 million questionnaires, received 2.3 million replies, and predicted Landon would beat Roosevelt. Identify the **two distinct** sources of error, explain the mechanism of each, and explain precisely why 2.3 million responses did not rescue the prediction. What does this say about the relation between sample size and bias?

2.4 [5 marks]

Write down the decomposition $total\ error = sampling\ error + non-sampling\ error$. Define each, give two examples of non-sampling error, and explain why increasing n can leave the total error unchanged or increase it.

2.5 [8 marks]

Define probability sampling. For each of *convenience*, *judgment*, *self-selection*, *snowball* and *quota* sampling give a one-line description, one situation where it is the pragmatic choice, and the specific bias it is most vulnerable to.

2.6 [8 marks]

Carefully distinguish a **sampling design** from a **sampling scheme**. Give an example of two genuinely different schemes inducing the *same* design, and explain why the design is what inference depends on.

3 Designs, selection and inclusion probabilities**3.1** [8 marks]

Let $U = \{1, 2, 3\}$ and $n = 2$. (a) Write down $\mathcal{S}$ and the SRSWOR design, and verify $\sum_s p(s) = 1$. (b) Compute π_j and $\pi_{jj'}$ directly and check against the general formulas. (c) Verify $\sum_j \pi_j = n$ and explain why this holds for *any* fixed-size design.

3.2 [10 marks]

(a) Show SRSWR induces the uniform design $p(\mathbf{s}) = 1/N^n$ on *ordered* samples. (b) Derive $\pi_j = 1 - (1 - \frac{1}{N})^n$. (c) Show $\pi_j < n/N$ for $n \geq 2$ and interpret the gap. (d) What is $\lim_{n \rightarrow \infty} \pi_j$, and what does this permit that SRSWOR does not?

3.3 [10 marks]

(a) Show SRSWOR induces the uniform design $p(\mathbf{s}) = 1/\binom{N}{n}$ on *unordered* samples. (b) Derive $\pi_j = n/N$ and $\pi_{jj'} = \frac{n(n-1)}{N(N-1)}$. (c) Verify $\sum_{j \neq j'} \pi_{jj'} = n(n-1)$.

3.4 [8 marks]

Under SRSWOR show that the *marginal* probability that unit j is selected at the k -th draw equals $1/N$ for every k . (*This is the “why?” on the Class-2 slide.*)

3.5 [5 marks]

With $N = 17$ and $n = 4$, describe the **rejection** and **remainder** methods for drawing an SRSWR from a table of random digits, carry out one draw of each from a stated row, and compare their rejection rates.

4 Estimation under SRSWR and SRSWOR**4.1** [10 marks]

Under SRSWR: (a) show $\bar{y}$ is unbiased for $\bar{Y}$; (b) show $\text{Var}(\bar{y}) = \sigma^2/n$; (c) state precisely

where the “with replacement” assumption is used in each part, and why $\text{Var}(\bar{y})$ does not depend on N except through σ^2 .

4.2 [8 marks]

Show $\mathbb{E}(s^2) = \sigma^2$ under SRSWR, and deduce that $\widehat{\text{Var}}(\bar{y}) = s^2/n$ is unbiased.

4.3 [8 marks]

For any known $h(\cdot)$, show $\frac{1}{n} \sum_i h(y_i)$ is unbiased for $\frac{1}{N} \sum_j h(Y_j)$ under SRSWR. Explain why this needs **no distributional assumption**, and what exactly makes it true.

4.4 [12 marks]

Under SRSWOR, with $y_1, \dots, y_n$ the values at the n draws: **(a)** Show $\mathbb{E}(y_i) = \bar{Y}$, $\text{Var}(y_i) = \sigma^2$. **(b)** Show $\text{Cov}(y_i, y_{i'}) = -\frac{\sigma^2}{N-1}$ for $i \neq i'$. **(c)** Derive $\text{Var}(\bar{y}) = \frac{N-n}{N-1} \cdot \frac{\sigma^2}{n} = (1-f) \frac{S^2}{n}$. **(d)**

Interpret the sign of the covariance and why it makes SRSWOR more efficient.

4.5 [10 marks]

Show $\mathbb{E}(s^2) = S^2$ under SRSWOR and deduce that $\widehat{\text{Var}}(\bar{y}) = (1-f)s^2/n$ is unbiased. (*Note the target is S^2 , not σ^2 .*)

4.6 [8 marks]

(a) Prove $\text{Var}_{\text{WOR}}(\bar{y}) < \text{Var}_{\text{WR}}(\bar{y})$ for $n \geq 2$, with equality iff $n = 1$. **(b)** Interpret the fpc $(1-f)$ and describe the regimes $f \rightarrow 0$ and $f \rightarrow 1$. **(c)** Give three reasons SRSWR is nonetheless used in practice.

4.7 [8 marks]

For the population total $T = \sum_j Y_j$ and $\hat{T} = N\bar{y}$: **(a)** Show $\hat{T}$ is unbiased and find $\text{Var}(\hat{T})$ under SRSWOR. **(b)** Give an unbiased estimator of $\text{Var}(\hat{T})$. **(c)** Show $\text{CV}(\hat{T}) = \text{CV}(\bar{y})$ and say why that is useful in a survey report.

5 Effective sample size and proportions

5.1 [10 marks]

Under SRSWR let n_{eff} be the number of *distinct* units sampled. **(a)** Explain why it is random. **(b)** Show $\mathbb{E}(n_{\text{eff}}) = N \left[1 - \left(1 - \frac{1}{N} \right)^n \right]$. **(c)** Evaluate for $N = 100$, $n = 20$ and comment. **(d)** Show $\mathbb{E}(n_{\text{eff}}) \leq n$ and say when the two are close.

5.2 [8 marks]

Let $\bar{y}_{\text{eff}}$ be the mean of Y over the distinct units of an SRSWR sample. Show $\mathbb{E}(\bar{y}_{\text{eff}}) = \bar{Y}$. (*Hint: condition on the set of distinct units and use symmetry.*) Compare $\bar{y}_{\text{eff}}$ with the SRSWOR mean of size n .

5.3 [10 marks]

Let $Y_j = \mathbb{I}(\text{unit } j \text{ has the characteristic})$ and P the population proportion. **(a)** Show $P = \bar{Y}$. **(b)** Show $\sigma^2 = P(1-P)$ and $S^2 = \frac{N}{N-1} P(1-P)$. **(c)** Write $\text{Var}(p)$ under SRSWR and SRSWOR. **(d)** Show $\widehat{\text{Var}}(p) = (1-f) \frac{p(1-p)}{n-1}$ is unbiased under SRSWOR.

5.4 [8 marks]

(a) Show $np \sim \text{Binomial}(n, P)$ under SRSWR, and state the distribution under SRSWOR. **(b)** Using these, identify the exact factor by which the SRSWOR variance is smaller.

6 Accuracy, confidence intervals, sample size

6.1 [5 marks]

Define the standard error and the coefficient of variation of an estimator, give $\widehat{SE}(\bar{y})$ and $\widehat{CV}(\bar{y})$ under SRSWOR, and state the Statistics Canada guidelines.

6.2 [10 marks]

An SRSWOR with $N = 5,000$, $n = 100$ gives $\bar{y} = \text{Rs. } 2,400$, $s = \text{Rs. } 1,800$. (a) Compute $\widehat{\text{Var}}(\bar{y})$, $\widehat{SE}(\bar{y})$, $\widehat{CV}(\bar{y})$. (b) Classify by the Statistics Canada guidelines. (c) Give an approximate 95% CI for $\bar{Y}$. (d) Estimate the total and give its estimated CV. (e) By what factor must n increase to halve the CV?

6.3 [10 marks]

(a) Explain why an *exact* 95% CI for $\bar{Y}$ cannot be computed in finite-population sampling, even though the sampling distribution of $\bar{y}$ is fully determined by the design. (b) State Hájek's (1960) result and its conditions. (c) Write the resulting approximate CI under SRSWOR and say when t_{n-1} should replace 1.96.

6.4 [8 marks]

A survey reports a 95% CI of [4735, 7265] for mean monthly rent, from $n = 15$. (a) State precisely what the coverage probability refers to. (b) A student says “there is a 95% probability that $\bar{Y}$ lies in this interval”. Explain why that is wrong here, and give the correct statement. (c) Comment on the interval's usefulness and on what should have been done instead.

6.5 [12 marks]

(a) From $\mathbb{P}(|\bar{y} - \bar{Y}| \leq e) = 1 - \alpha$ and the normal approximation, derive $n_{WR} = \frac{z_{\alpha/2}^2 \sigma^2}{e^2}$. (b) Derive $n_{WOR} = \frac{z_{\alpha/2}^2 S^2}{e^2 + z_{\alpha/2}^2 S^2 / N}$. (c) Hence show $n_{WOR} = \frac{n_{WR}}{1 + n_{WR}/N}$, and deduce $n_{WOR} < N$ always. (d) Comment on the claim that one should sample “a fixed percentage of the population”.

6.6 [10 marks]

(a) Derive $n_{WR} = \frac{z_{\alpha/2}^2 P(1-P)}{e^2}$ for a proportion. (b) Show $P(1-P) \leq \frac{1}{4}$ and obtain the worst-case n . (c) At 95% confidence compute the worst-case n for $e = 0.10, 0.05, 0.03$ and comment on the scaling. (d) If in truth $P = 0.2$, quantify how wasteful the worst-case choice was.

6.7 [8 marks]

(a) Show that with a *relative* margin of error r , the required n depends on the population **coefficient of variation**. Derive the formula. (b) Explain why the CV is easier to guess in advance than σ^2 .

6.8 [12 marks]

For $H_0 : \bar{Y} = \mu_0$ against $H_1 : \bar{Y} = \mu_1 > \mu_0$, with $\Delta = \mu_1 - \mu_0$: (a) Write the test statistic and level- α rejection rule, stating the approximation used. (b) Derive $n_{WR} = \frac{(z_{\alpha} + z_{\beta})^2 \sigma^2}{\Delta^2}$ from the power requirement. (c) State the two-sided and SRSWOR versions. (d) Explain the role of the *minimum detectable effect* and why Δ is a subject-matter choice.

6.9 [10 marks]

Among people free of diabetes, mean fasting glucose is 95.0 mg/dL with sd 9.8; a mean of 100 in coffee drinkers would be clinically important. Under SRSWR, how many patients give a two-sided 5% test power 80%? Show all working and state assumptions. ($z_{0.025} = 1.96$, $z_{0.20} = 0.8416$.)

6.10

[10 marks]

26% of people free of heart disease have elevated LDL. How many patients are needed for a two-sided 5% test to have power 90% to detect a difference of 0.05 in the proportion? ($z_{0.025} = 1.96$, $z_{0.10} = 1.2816$.)

6.11

[8 marks]

Your required n is three times the budget. Describe three distinct ways to reduce it, state exactly what each costs you, and say which you would choose and why.

7 Stratified random sampling

7.1

[8 marks]

A campus health centre wants the mean height of 1,000 undergraduates, half male and half female. An SRSWOR of size 10 happens to return 9 men and 1 woman. (a) Is the estimator $\bar{y}$ biased? Answer precisely, and explain the difference between being representative *in expectation* and being representative *in this sample*. (b) Explain how stratification fixes the problem, and what extra information it requires. (c) Is a stratified sample still a probability sample? Justify.

7.2

[12 marks]

Set up stratified random sampling with H strata, N_h , n_h , and SRSWOR drawn **independently** within each stratum. (a) Show that $\bar{y} = \frac{1}{n} \sum$ (all sampled values) is in general *biased* for $\bar{Y}$, and state exactly when it is unbiased. (b) Show that $\bar{y}_{st} = \frac{1}{N} \sum_h N_h \bar{y}_h$ is unbiased. (c) Derive $\text{Var}(\bar{y}_{st}) = \frac{1}{N^2} \sum_h N_h^2 (1 - f_h) \frac{S_h^2}{n_h}$, stating clearly where the independence across strata is used. (d) Write down an unbiased estimator of this variance and the approximate 95% CI for $\bar{Y}$.

7.3

[10 marks]

(a) Define equal, proportional and Neyman allocation, and state the one-line rationale for each. (b) Show that under **proportional** allocation $f_h = f$ for every h , and hence $\text{Var}_{prop}(\bar{y}_{st}) = \frac{1-f}{n} \sum_h W_h S_h^2$. (c) Show that under **Neyman** allocation $\text{Var}_{Ney}(\bar{y}_{st}) = \frac{1}{n} (\sum_h W_h S_h)^2 - \frac{1}{N} \sum_h W_h S_h^2$. (d) Deduce $\text{Var}_{prop} \geq \text{Var}_{Ney}$, and state exactly when equality holds.

7.4

[10 marks]

Prove that Neyman allocation minimises $\text{Var}(\bar{y}_{st})$ subject to $\sum_h n_h = n$. (Use a Lagrange multiplier, or the Cauchy-Schwarz inequality.)

7.5

[10 marks]

Suppose the cost of the survey is $A + \sum_h n_h C_h$, where A is overhead and C_h the cost per unit in stratum h . (a) Formulate the problem of minimising $\text{Var}(\bar{y}_{st})$ subject to a total budget C . (b) Show the optimum is $n_h \propto \frac{N_h S_h}{\sqrt{C_h}}$, and write n_h explicitly. (c) Show this reduces to Neyman allocation when C_h is constant, and explain intuitively why expensive strata get smaller samples.

7.6

[12 marks]

Compare an SRSWOR of size n with a stratified WOR sample of the same size n under proportional allocation. The exact identity is

$$\text{Var}_{SRS}(\bar{y}) - \text{Var}_{st}(\bar{y}_{st}) = \frac{N-n}{n(N-1)} \left[\sum_{h=1}^H W_h (\bar{Y}_h - \bar{Y})^2 - \frac{1}{N} \sum_{h=1}^H (1 - W_h) S_h^2 \right].$$

(a) Identify the two bracketed terms as between-stratum and within-stratum contributions. (b)

State the condition under which stratification helps, in words. (c) What happens in the special case $\bar{Y}_1 = \dots = \bar{Y}_H$? What has gone wrong, and what does it tell you about how to form strata? (d) Why is stratification nevertheless “almost always beneficial” in realistic surveys?

7.7

[12 marks]

A district has $N = 1,000$ households in three strata: urban ($N_1 = 200$), semi-urban ($N_2 = 300$), rural ($N_3 = 500$). From a pilot, $S_1 = 60$, $S_2 = 40$, $S_3 = 20$ and $\bar{Y}_1 = 300$, $\bar{Y}_2 = 200$, $\bar{Y}_3 = 100$ (rupees per month). Take $n = 100$. (a) Compute the proportional and the Neyman allocations. (b) Compute $\text{Var}_{prop}(\bar{y}_{st})$ and $\text{Var}_{Ney}(\bar{y}_{st})$. (c) Compute $\bar{Y}$ and (using $S^2 \approx \sum W_h S_h^2 + \sum W_h (\bar{Y}_h - \bar{Y})^2$) the variance under SRSWOR of the same size. Compare all three and comment. (d) Neyman allocation gives non-integer n_h . How would you round, and what problem does rounding create?

7.8

[8 marks]

Give four practical reasons, beyond variance reduction, for using stratified sampling. For each, give a concrete example. (Class 8, “Stratification: some remarks”.)

8 Systematic sampling

8.1

[10 marks]

Let $N = nk$ with k an integer. (a) State the linear systematic sampling scheme precisely. (b) Write down $\mathcal{S}$, the sampling design, and show $\pi_j = n/N$ for every j . (c) Compute $\pi_{jj'}$ and show it is 0 for some pairs. Explain why this single fact is responsible for the variance-estimation problem of Class 9.

8.2

[12 marks]

With $N = nk$, write Y_{ij} for the j -th unit of cluster i , $i = 1, \dots, k$. (a) Show that a systematic sample is exactly one randomly chosen cluster, and hence a special case of cluster sampling. (b) Show $\mathbb{E}(\bar{y}) = \bar{Y}$. (c) Show $\text{Var}(\bar{y}) = \frac{1}{k} \sum_{i=1}^k (\bar{Y}_i - \bar{Y})^2 = \sigma_b^2$, the **between-cluster** variance. (d) Deduce the design rule: for systematic sampling to be efficient, should clusters be internally homogeneous or heterogeneous? Contrast with stratified sampling.

8.3

[10 marks]

Let ρ_c be the intraclass correlation coefficient. (a) Using $\text{Var}(\bar{y}) = \frac{\sigma^2}{n} \{1 + (n-1)\rho_c\}$, show that systematic sampling beats SRSWR iff $\rho_c \leq 0$. (b) Show that it beats SRSWOR iff $\rho_c \leq -\frac{1}{N-1}$. (c) Show $-\frac{1}{n-1} \leq \rho_c \leq 1$ and interpret both extremes.

8.4

[12 marks]

Take $N = 12$, $n = 3$, $k = 4$. (a) Population A has $Y = (1, 2, \dots, 12)$ in that order. List the four possible systematic samples, compute $\text{Var}(\bar{y})$ directly, and compare with the SRSWR and SRSWOR variances. (b) Population B has $Y = (10, 20, 30, 40)$ repeated three times. Repeat the computation. (c) Compute ρ_c in each case from $\text{Var}(\bar{y}) = \frac{\sigma^2}{n} \{1 + (n-1)\rho_c\}$, and explain the two results in terms of the ordering of the sampling frame.

8.5

[10 marks]

(a) Explain why $\text{Var}(\bar{y})$ cannot be estimated unbiasedly from a single systematic sample. (b) Suppose we pretend the sample is SRSWOR and use $\hat{V} = (1-f)\frac{s^2}{n}$. Given that $\mathbb{E}(\hat{V}) = (1-f)\frac{S^2}{n}$.

$\frac{N-1}{N}(1 - \rho_c)$, determine the direction of the bias as a function of ρ_c , and state when $\hat{V}$ is unbiased.
 (c) When is this pretence a *conservative* (safe) thing to do?

8.6

[10 marks]

Describe three practical ways of obtaining a usable variance estimate under systematic sampling, and state the cost or assumption each one carries: (i) m independent systematic samples (IPNS); (ii) m random starts within the first mk units; (iii) the successive-difference estimator of Wolter (1984). Write down the estimator in each case.

8.7

[8 marks]

The ISI canteen has $N = 14$ staff and you want $n = 4$. (a) Show that linear systematic sampling breaks down, and describe the two ad hoc fixes with $k = 3$ and $k = 4$, listing the resulting samples and sizes. (b) Describe **circular systematic sampling** and carry it out with $k = 4$ and random start $i = 13$. (c) Show that circular systematic sampling gives $\pi_j = n/N$ for every j , and explain why this is an improvement on the ad hoc fixes.

8.8

[8 marks]

State four situations in which systematic sampling is preferred to SRS in practice, and two situations in which it can go badly wrong. Give a concrete example of each failure.

9 Applied and mixed problems

9.1

[12 marks]

A district has $N = 1,200$ registered small businesses. An SRSWOR of $n = 150$ finds that 42 rely on local moneylenders rather than bank loans. (a) Estimate P and give the estimated standard error. (b) Give an approximate 95% CI for P . (c) Estimate the total number of such businesses, with its standard error. (d) Compute the estimated CV and classify it. (e) A follow-up wants margin of error 0.03 at 95%. Find the required n (i) using the worst-case bound and (ii) using this survey as a pilot.

9.2

[10 marks]

Critically evaluate the following design, identifying every flaw and classifying each as a sampling or a non-sampling error.

To estimate the average monthly household expenditure on private tuition in a city, a researcher stands outside three well-known coaching centres on a weekday evening and interviews the first 200 parents who agree to talk. Of the 200 approached, 120 respond. The questionnaire asks "How much do you spend on your child's tuition?" The researcher reports the sample mean as the city average, with standard error $s/\sqrt{120}$.

9.3

[10 marks]

A population of $N = 6$ units has $Y = (2, 4, 4, 6, 8, 12)$. (a) Compute $\bar{Y}$, σ^2 , S^2 and verify $S^2 = \frac{N}{N-1}\sigma^2$. (b) For $n = 2$ under SRSWOR, list all $\binom{6}{2}$ samples, verify $\mathbb{E}(\bar{y}) = \bar{Y}$ directly and check $\text{Var}(\bar{y})$ against $(1 - f)S^2/n$. (c) Compute $\text{Var}_{WR}(\bar{y})$ and confirm the inequality.

9.4

[12 marks]

You must design a survey to estimate mean household electricity consumption in a city of $N = 100,000$ households. You know the ward of every household, and previous studies suggest consumption varies far more *between* wards than *within* them. The budget allows $n = 2,000$. (a) Which of SRSWOR, stratified and systematic sampling would you choose, and why? Address all three. (b) If you stratify by ward, which allocation would you use, and what would you need to

know? (c) Suppose the household list is sorted by ward. What does a systematic sample of every 50th household resemble, and is that good or bad here? (d) State one non-sampling error this survey is particularly exposed to, and how you would mitigate it.

10 Solutions: concepts and vocabulary

10.1

[5 marks]

Solution.

- **Observation unit** — the object measured. Here: an adult person.
- **Target population** — the ideal complete collection we want to infer about, in a stated period. Here: all adults in West Bengal.
- **Sampled population** — the accessible part, i.e. the units that *could* be chosen. Here: adults in registered households.
- **Sampling unit** — a unit that can be selected. Need not equal the observation unit: with no list of adults, the sampling unit could be the *household*.
- **Sampling frame** — the tangible list/map/database of sampling units with unique IDs. Here: the latest West Bengal Electoral Roll.

If sampled $\neq$ target: inference is valid only for the sampled population. The difference is a **coverage error**, a bias that no increase in n removes, because excluded units have zero probability of selection.

10.2

[5 marks]

Solution. A **census** measures every unit; a **sample survey** measures a subset and infers. Three reasons a survey can be more accurate:

1. **Better fieldwork** — a small sample can be handled by few, highly trained investigators, so measurement error per unit is lower.
2. **Enlarged scope** — with fewer units you can ask more and probe harder within the same budget.
3. **Speed** — a census takes so long the population changes meanwhile.

A census removes *sampling* error but typically has much larger *non-sampling* error, and only the total matters.

10.3

[8 marks]

Solution. Error 1 — selection bias (frame problem). Names came from telephone directories and vehicle registrations, which in 1936 over-represented richer, Republican-leaning households and excluded poorer, Roosevelt-leaning ones. The sampled population was not the target population of voters.

Error 2 — non-response bias. Only 2.3 of 10 million replied, and responding was correlated with preference: Landon supporters were far more motivated to return the form.

Why 2.3 million did not help. Both are **biases**, not variances: $MSE = \text{Bias}^2 + \text{Variance}$. Increasing n drives the variance to zero but leaves the bias untouched, producing a very *precise* estimate of the *wrong* quantity.

Moral: sample size buys precision; only the design buys unbiasedness. A small probability sample beats a huge convenience sample.

10.4

[5 marks]

Solution.

$$\text{Total error} = \underbrace{\text{Sampling error}}_{\text{from observing only part of } U} + \underbrace{\text{Non-sampling error}}_{\text{everything else}}.$$

Sampling error is what $\text{Var}(\bar{y})$ measures and $\rightarrow 0$ as $n \rightarrow N$. Non-sampling error includes measurement error (bad or sensitive questions), non-response, frame/coverage error and poorly trained investigators.

Only the first term shrinks with n . Worse, a larger survey usually needs more, less well trained investigators and suffers lower response rates, so the second term *grows*. This is why “bigger is always better” is false.

10.5**[8 marks]**

Solution. Probability sampling: selection is governed by a randomisation mechanism giving every unit in the frame a **known, strictly positive** probability of selection. That known mechanism is what makes design-based inference possible.

Method	Description	Typical use	Main vulnerability
Convenience	take whoever is accessible and cheap	pilot test of wording	gross selection bias
Judgment	researcher hand-picks informative units	expert interviews, case studies	the researcher's prior
Self-selection	individuals opt in	feedback forms, internet polls	only the opinionated respond
Snowball	selected units refer others	hidden populations with no frame	network homophily
Quota	fill target counts per group by convenience	guaranteed demographic mix, cheap	convenience bias <i>within</i> quotas

All five share one fatal defect: selection probabilities are unknown, so no design-based standard error or CI can be computed.

10.6**[8 marks]**

Solution. A **sampling design** is a probability distribution p over $\mathcal{S}$ with $\sum_s p(s) = 1$ — the mathematical object. A **sampling scheme** is an algorithm for drawing a sample — the operational recipe.

Two schemes, one design. For SRSWOR of size n : *Scheme A* — draw units one at a time, each remaining unit equally likely, then discard the order. *Scheme B* — enumerate all $\binom{N}{n}$ subsets and pick one uniformly. Scheme A gives each ordered sequence probability $\frac{(N-n)!}{N!}$ and each unordered set arises from $n!$ orderings, so $p(\mathbf{s}) = \frac{n!(N-n)!}{N!} = 1/\binom{N}{n}$ — identical to B.

Why the design matters. $\mathbb{E}(\bar{y})$, $\text{Var}(\bar{y})$, π_j , $\pi_{jj'}$ are functionals of $p(\cdot)$ alone. Two schemes with the same design are statistically indistinguishable; the choice between them is purely practical. Conversely a scheme is useless if you cannot write down the design it induces.

11 Solutions: designs and inclusion probabilities**11.1****[8 marks]**

(a) $\mathcal{S} = \{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$, $p(s) = 1/\binom{3}{2} = \frac{1}{3}$ each; sum = 1 ✓

(b) Unit 1 lies in two samples, so $\pi_1 = \frac{2}{3}$; by symmetry all $\pi_j = \frac{2}{3} = n/N$ ✓. Each pair lies in exactly one sample, so $\pi_{jj'} = \frac{1}{3} = \frac{2 \cdot 1}{3 \cdot 2}$ ✓

(c) $\sum_j \pi_j = 3 \cdot \frac{2}{3} = 2 = n$ ✓. In general, with I_j the inclusion indicator,

$$\sum_j \pi_j = \sum_j \mathbb{E}(I_j) = \mathbb{E}\left(\sum_j I_j\right) = \mathbb{E}(\text{sample size}) = n,$$

for *any* fixed-size design.

11.2

[10 marks]

- (a) An ordered sample is $(u_1, \dots, u_n)$; each draw independently picks any unit with probability $1/N$, so $p(\mathbf{s}) = 1/N^n$ for all N^n ordered samples — uniform.
- (b) Unit j is excluded iff all n independent draws miss it: $\mathbb{P}(j \notin \mathbf{s}) = (1 - \frac{1}{N})^n$, so $\pi_j = 1 - (1 - \frac{1}{N})^n$.
- (c) By the binomial expansion, $(1 - \frac{1}{N})^n > 1 - \frac{n}{N}$ strictly for $n \geq 2$, hence $\pi_j < n/N$. **Interpretation:** n/N is the SRSWOR inclusion probability; under SRSWR some draws are wasted on repeats, so a given unit is less likely to appear at all.
- (d) $\pi_j \rightarrow 1$. Also n may exceed N under SRSWR — impossible under SRSWOR — which is exactly what makes SRSWR the right tool for the bootstrap.

11.3

[10 marks]

- (a) A particular ordered sequence of n distinct units has probability $\frac{1}{N} \cdot \frac{1}{N-1} \cdots \frac{1}{N-n+1} = \frac{(N-n)!}{N!}$. Each unordered sample arises from $n!$ orderings, so $p(\mathbf{s}) = n! \frac{(N-n)!}{N!} = 1/\binom{N}{n}$ — uniform.
- (b) Counting samples, $\pi_j = \binom{N-1}{n-1} / \binom{N}{n} = \frac{n}{N}$ and $\pi_{jj'} = \binom{N-2}{n-2} / \binom{N}{n} = \frac{n(n-1)}{N(N-1)}$.
- (c) There are $N(N-1)$ ordered pairs, so $\sum_{j \neq j'} \pi_{jj'} = N(N-1) \cdot \frac{n(n-1)}{N(N-1)} = n(n-1)$ ✓ (Sanity: $\mathbb{E}[(\sum_j I_j)^2] = n + n(n-1) = n^2$, as required for a fixed-size design.)

11.4

[8 marks]

Solution. Unit j is selected at draw k iff it survives the first $k-1$ draws and is then chosen:

$$\mathbb{P} = \underbrace{\frac{N-1}{N} \cdot \frac{N-2}{N-1} \cdots \frac{N-k+1}{N-k+2}}_{\text{telescopes to } \frac{N-k+1}{N}} \times \frac{1}{N-k+1} = \frac{1}{N},$$

independent of k . ■ **This is the “S” in SRS:** equal marginal selection probability at *every* stage, even though the conditional probabilities $\frac{1}{N-k+1}$ change.

11.5

[5 marks]

Solution. Rejection method. Read two-digit numbers; accept if in 01–17, reject if > 17 or $= 00$. Only 17% of numbers are usable — about 83% wasted.

Remainder method. The largest two-digit multiple of 17 is $17 \times 5 = 85$. Reject only if > 85 or $= 00$; otherwise take $x \bmod 17$ (remainder 0 $\mapsto$ unit 17). Only 14% wasted. Each residue class receives exactly five of the 85 admissible numbers, so selection remains exactly uniform — which is why trimming to 85 is essential and cannot be skipped.

From row 07 (77783 00015 10806 ...): rejection gives 15, 10, ...; remainder gives 17, 10, 13, 15.

12 Solutions: estimation under SRSWR and SRSWOR

12.1

[10 marks]

- (a) Each draw is uniform on U , so $\mathbb{E}(y_i) = \sum_j \frac{1}{N} Y_j = \bar{Y}$, and $\mathbb{E}(\bar{y}) = \frac{1}{n} \sum_i \mathbb{E}(y_i) = \bar{Y}$. ■
- (b) $\text{Var}(y_i) = \frac{1}{N} \sum_j Y_j^2 - \bar{Y}^2 = \sigma^2$. Because draws are **with replacement**, $y_1, \dots, y_n$ are *independent*, so $\text{Var}(\bar{y}) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}$. ■
- (c) The WR assumption is *not* used in (a) — unbiasedness needs only the marginal distribution of each draw and holds for SRSWOR too. It is essential in (b), to kill the covariances. N does not appear because each draw is an independent uniform pick from the same distribution, exactly like iid sampling from an infinite population putting mass $1/N$ on each Y_j ; N enters only through the shape of that distribution, i.e. through σ^2 .

12.2

[8 marks]

Solution. With $s^2 = \frac{1}{n-1}(\sum_i y_i^2 - n\bar{y}^2)$,

$$\mathbb{E}\left(\sum_i y_i^2\right) = n(\sigma^2 + \bar{Y}^2), \quad \mathbb{E}(\bar{y}^2) = \frac{\sigma^2}{n} + \bar{Y}^2,$$

so $\mathbb{E}(s^2) = \frac{1}{n-1}[n\sigma^2 - \sigma^2] = \sigma^2$.

■ Hence $\mathbb{E}(s^2/n) = \sigma^2/n = \text{Var}(\bar{y})$.

Trap — the divisor

s^2 (divisor $n-1$) is unbiased for σ^2 (divisor N) under SRSWR, but for S^2 (divisor $N-1$) under SRSWOR. **Same statistic, different target.**

12.3

[8 marks]

Solution. Since y_i is a uniform draw from $\{Y_1, \dots, Y_N\}$, $\mathbb{E}(h(y_i)) = \frac{1}{N} \sum_j h(Y_j)$; averaging over i gives the result. ■

Why no distributional assumption is needed. The Y_j are **fixed** numbers — there is no population distribution to assume anything about. The only randomness is the randomness *we introduced* when choosing units, and its distribution is known exactly because we designed it. Unbiasedness therefore holds **by design**, for every conceivable population: *the randomisation is the assumption*.

Taking $h(y) = y$ gives the mean; $h(y) = y^2$ gives $\frac{1}{N} \sum Y_j^2$; $h(y) = \mathbb{I}(y \leq t)$ gives the population distribution function at t .

12.4

[12 marks]

(a) By Question 3.4 the unit drawn at any single stage is uniform on U , so the marginals are as in the WR case: $\mathbb{E}(y_i) = \bar{Y}$, $\text{Var}(y_i) = \sigma^2$; hence $\mathbb{E}(\bar{y}) = \bar{Y}$ — the sample mean is unbiased under SRSWOR too.

(b) For $i \neq i'$ the pair is uniform over the $N(N-1)$ ordered pairs of distinct units:

$$\mathbb{E}(y_i y_{i'}) = \frac{1}{N(N-1)} \left[\left(\sum_j Y_j \right)^2 - \sum_j Y_j^2 \right] = \frac{N^2 \bar{Y}^2 - N(\sigma^2 + \bar{Y}^2)}{N(N-1)} = \bar{Y}^2 - \frac{\sigma^2}{N-1},$$

so $\text{Cov}(y_i, y_{i'}) = -\frac{\sigma^2}{N-1}$. ■

(c) With n variance terms and $n(n-1)$ ordered covariance terms,

$$\text{Var}(\bar{y}) = \frac{1}{n^2} \left[n\sigma^2 - n(n-1) \frac{\sigma^2}{N-1} \right] = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}.$$

Since $S^2 = \frac{N}{N-1} \sigma^2$, i.e. $\sigma^2 = \frac{N-1}{N} S^2$,

$$\text{Var}(\bar{y}) = \frac{N-1}{N} \cdot \frac{S^2}{n} \cdot \frac{N-n}{N-1} = \frac{N-n}{N} \cdot \frac{S^2}{n} = (1-f) \frac{S^2}{n}. \quad \square$$

(d) The covariance is **negative** because the draws compete: a large value taken at draw i is removed from the pool, making draw i' slightly more likely to be small. This negative dependence spreads the sample more evenly over the population, so $\bar{y}$ varies less — exactly the reduction expressed by $\frac{N-n}{N-1} < 1$.

12.5

[10 marks]

Solution. As before, $\mathbb{E}(s^2) = \frac{1}{n-1} [n\sigma^2 - (1-f)S^2]$. Substituting $\sigma^2 = \frac{N-1}{N} S^2$ and $1-f = \frac{N-n}{N}$,

$$n\sigma^2 - (1-f)S^2 = \frac{S^2}{N} [n(N-1) - (N-n)] = \frac{S^2}{N} \cdot N(n-1) = (n-1)S^2,$$

so $\mathbb{E}(s^2) = S^2$.

■ Hence $\mathbb{E}[(1-f)s^2/n] = (1-f)S^2/n = \text{Var}(\bar{y})$.

12.6

[8 marks]

(a) $\frac{\text{Var}_{WOR}}{\text{Var}_{WR}} = \frac{N-n}{N-1} < 1 \iff n > 1$; equality at $n = 1$, as it must be since one draw is identical under the two schemes. ■

(b) $(1-f)$ is the **finite population correction** — the price SRSWR pays for allowing repeats. $f \rightarrow 0$: $(1-f) \rightarrow 1$ and the schemes agree (why national surveys ignore the distinction). $f \rightarrow 1$: $\text{Var}(\bar{y}) \rightarrow 0$ — a census has no sampling error. The WR variance σ^2/n never does this, a good check that the fpc must be present.

(c) (i) For large N the difference is negligible and the bookkeeping is simpler; (ii) it saves computation and memory — no need to track what was drawn; (iii) **bootstrap** methods resample with replacement, and there n may legitimately exceed the number of distinct units.

12.7

[8 marks]

(a) $\mathbb{E}(\hat{T}) = N\bar{Y} = T$ and $\text{Var}(\hat{T}) = N^2(1-f)\frac{S^2}{n} = N(N-n)\frac{S^2}{n}$.

(b) $\widehat{\text{Var}}(\hat{T}) = N(N-n)\frac{s^2}{n}$, unbiased since $\mathbb{E}(s^2) = S^2$.

(c) $\text{CV}(\hat{T}) = \frac{N \text{SE}(\bar{y})}{N\bar{Y}} = \text{CV}(\bar{y})$. **Useful** because the CV is scale-free: one reliability statement covers the mean, the total, and anything obtained by multiplying by a known constant, so a report can quote one CV and apply the Statistics Canada classification to all of them.

13 Solutions: effective sample size and proportions

13.1

[10 marks]

(a) Under SRSWR a unit may be drawn more than once; how many distinct units appear depends on which repeat, and that is determined by the random draws.

(b) Write $n_{\text{eff}} = \sum_j I_j$ with $I_j = \mathbb{I}(j \text{ appears at least once})$. By Question 3.2(b), $\mathbb{E}(I_j) = 1 - (1 - \frac{1}{N})^n$, so by linearity (no independence needed) $\mathbb{E}(n_{\text{eff}}) = N[1 - (1 - \frac{1}{N})^n]$. ■

(c) $N = 100, n = 20$: $(0.99)^{20} = 0.8179$, so $\mathbb{E}(n_{\text{eff}}) = 18.21$ — about 1.8 of the 20 draws wasted, a 9% loss.

(d) $\pi_j \leq n/N$ gives $\mathbb{E}(n_{\text{eff}}) = N\pi_j \leq n$. For $n \ll N$, $\mathbb{E}(n_{\text{eff}}) \approx n - \frac{n(n-1)}{2N}$, so the loss is negligible when $n^2 \ll N$ — the birthday-problem threshold.

13.2

[8 marks]

Solution. Let D be the random set of distinct units and condition on $|D| = m$.

Symmetry. The SRSWR design is symmetric in the unit labels (every ordered sample has probability $1/N^n$), so relabelling by any permutation leaves the distribution of D unchanged. Hence given $|D| = m$, D is **uniform over all** $\binom{N}{m}$ **subsets of size** m .

Conditional expectation. Given $|D| = m$, D is an SRSWOR sample of size m , so $\mathbb{E}(\bar{y}_{\text{eff}} | |D| = m) = \bar{Y}$ for every m .

Tower property. The conditional expectation is the constant $\bar{Y}$, so $\mathbb{E}(\bar{y}_{\text{eff}}) = \bar{Y}$. ■

Comparison. $\bar{y}_{\text{eff}}$ is an SRSWOR mean of *random* size $n_{\text{eff}} \leq n$, so it cannot beat a genuine SRSWOR of size n ; but it does beat the ordinary SRSWR mean, since it discards the redundant repeats: $\text{Var}_{WOR,n} \leq \text{Var}(\bar{y}_{\text{eff}}) \leq \text{Var}_{WR,n}$.

13.3

[10 marks]

(a) $\bar{Y} = \frac{1}{N} \sum_j Y_j = \frac{\#\{j: Y_j=1\}}{N} = P$. A proportion is a mean, so every earlier result applies and $p = \bar{y}$.

(b) $Y_j^2 = Y_j$, so $\sum_j Y_j^2 = NP$ and $\sigma^2 = P - P^2 = P(1-P)$, $S^2 = \frac{N}{N-1}P(1-P)$.

$$(c) \text{Var}_{WR}(p) = \frac{P(1-P)}{n}; \text{Var}_{WOR}(p) = (1-f) \frac{S^2}{n} = \frac{N-n}{N-1} \cdot \frac{P(1-P)}{n}.$$

(d) For binary data $\bar{y} = p$, $y_i^2 = y_i$, so $s^2 = \frac{np - np^2}{n-1} = \frac{np(1-p)}{n-1}$ and $\widehat{\text{Var}}(p) = (1-f) \frac{s^2}{n} = (1-f) \frac{p(1-p)}{n-1}$, unbiased since $\mathbb{E}(s^2) = S^2$. ■

Trap — $n-1$, not n

$\frac{p(1-p)}{n}$ is the iid formula and is **biased** here. The finite-population estimator needs $n-1$ and the factor $(1-f)$. Both corrections carry marks.

13.4

[8 marks]

(a) Under SRSWR the draws are independent Bernoulli(P), so $np \sim \text{Binomial}(n, P)$ and $\text{Var}(p) = P(1-P)/n$. Under SRSWOR the sample is a uniform n -subset of a population with NP ones, so $np \sim \text{Hypergeometric}(N, NP, n)$.

(b) The hypergeometric variance is $nP(1-P) \frac{N-n}{N-1}$, so

$$\text{Var}_{WOR}(p) = \text{Var}_{WR}(p) \times \frac{N-n}{N-1}.$$

The factor is $\frac{N-n}{N-1}$ — the same fpc as for a general Y , which is no coincidence: the binary case is a special case, not a separate theory.

14 Solutions: accuracy, CIs, sample size

14.1

[5 marks]

Solution. $\text{SE}(\bar{y}) = \sqrt{\text{Var}(\bar{y})}$ measures sampling variability on the data scale; $\text{CV}(\bar{y}) = \text{SE}(\bar{y})/\bar{Y}$ measures it relative to the truth and is scale-free. Under SRSWOR, $\widehat{\text{SE}}(\bar{y}) = \sqrt{1-f} \frac{s}{\sqrt{n}}$ and $\widehat{\text{CV}}(\bar{y}) = \widehat{\text{SE}}(\bar{y})/\bar{y}$.

Statistics Canada: $\text{CV} \leq 16.6\%$ reliable; $16.6\% < \text{CV} \leq 33.3\%$ publish with a warning; $\text{CV} > 33.3\%$ unreliable.

14.2

[10 marks]

Solution. $f = 100/5000 = 0.02$. (a) $\widehat{\text{Var}}(\bar{y}) = 0.98 \times \frac{1800^2}{100} = 31,752$; $\widehat{\text{SE}} = 178.19$; $\widehat{\text{CV}} = 178.19/2400 = 7.42\%$. (b) $7.42\% \leq 16.6\%$: **reliable**. (c) $2400 \pm 1.96(178.19) = 2400 \pm 349.3$, i.e. [Rs. 2051, Rs. 2749]. (d) $\hat{T} = 12,000,000$ with $\widehat{\text{SE}} = 890,950$ and $\widehat{\text{CV}} = 7.42\%$ — identical, as Question 4.7(c) requires. (e) Halving the CV quarters the variance: $(1 - \frac{n}{5000}) \frac{1800^2}{n} = 7938 \Rightarrow \frac{3,240,000}{n} = 7938 \Rightarrow n = 377.4$. So $n \approx 378$, a factor of 3.8 — slightly less than the naive $4\times$ because the fpc helps as n grows. Halving the error costs nearly four times the fieldwork.

14.3

[10 marks]

(a) The sampling distribution of $\bar{y}$ is determined by the design and by $Y_1, \dots, Y_N$. To compute an exact quantile of $\bar{y} - \bar{Y}$ we would need those values — but if we knew them we would know $\bar{Y}$ and would not be sampling. The distribution is also discrete on the $\binom{N}{n}$ sample means with no closed form.

(b) **Hájek (1960).** Under regularity conditions on the finite population (a Lindeberg-type condition ensuring no Y_j dominates), and provided N, n and $N-n$ are all large, $\frac{\bar{y} - \bar{Y}}{\widehat{\text{SE}}(\bar{y})} \stackrel{\text{approx}}{\sim} \mathcal{N}(0, 1)$. The requirement that $N-n$ be large is distinctive to finite populations — as $n \rightarrow N$ the estimator degenerates.

(c) $[\bar{y} \pm 1.96 \widehat{SE}(\bar{y})]$ with $\widehat{SE}(\bar{y}) = \sqrt{1-f} \frac{s}{\sqrt{n}}$. For **small** n the extra variability from estimating SE matters, and $t_{n-1,0.975}$ should replace 1.96.

14.4

[8 marks]

(a) 0.95 is a property of the **procedure**: over repeated samples from the same population, about 95% of the random intervals so constructed contain $\bar{Y}$.

(b) $\bar{Y}$ is a **fixed unknown number**; the only randomness is the sampling. Once the interval is computed it either contains $\bar{Y}$ or not — probability 0 or 1. The correct statement is (a). (A genuine probability statement about $\bar{Y}$ needs a Bayesian prior, which was not used.)

(c) 6000 ± 1265 is a margin of error over 21% of the estimate — not actionable. With $n = 15$ the normal approximation is doubtful and t_{14} should have been used, widening it further. **What to do instead**: fix the tolerable margin of error first (say $e = 300$) and solve for the n that delivers it.

14.5

[12 marks]

(a) By the normal approximation, matching $\mathbb{P}(|\bar{y} - \bar{Y}| \leq z_{\alpha/2} \text{SE}) \approx 1 - \alpha$ to the requirement gives $e = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$, hence $n_{WR} = \frac{z_{\alpha/2}^2 \sigma^2}{e^2}$.

(b) Under SRSWOR $\text{SE}^2 = \frac{S^2}{n} - \frac{S^2}{N}$, so $\frac{e^2}{z^2} = \frac{S^2}{n} - \frac{S^2}{N}$, giving $n_{WOR} = \frac{z^2 S^2}{e^2 + z^2 S^2 / N}$.

(c) Dividing through by e^2 and writing $n_{WR} = z^2 S^2 / e^2$: $n_{WOR} = \frac{n_{WR}}{1 + n_{WR}/N} < n_{WR}$. Also $n_{WOR} = \frac{N n_{WR}}{N + n_{WR}} < N$ — the required sample never exceeds the population, which the WR formula fails to guarantee.

(d) The claim is wrong: n_{WR} **does not involve** N **at all**; N enters only through $1/(1 + n_{WR}/N)$, which *reduces* the requirement for small populations. A city of 10^5 and a country of 10^9 need essentially the same sample for the same precision — which is why national polls of ~ 1000 work.

14.6

[10 marks]

(a) Put $\sigma^2 = P(1 - P)$: $n_{WR} = \frac{z_{\alpha/2}^2 P(1-P)}{e^2}$. (b) $P(1 - P) = \frac{1}{4} - (P - \frac{1}{2})^2 \leq \frac{1}{4}$, equality iff $P = \frac{1}{2}$; so $n_{WR} \leq \frac{z_{\alpha/2}^2}{4e^2}$, achievable without knowing P . (c) With $z^2/4 = 0.9604$:

e	$0.9604/e^2$	round up
0.10	96.04	97
0.05	384.16	385
0.03	1067.1	1068

$n \propto 1/e^2$: halving the margin **quadruples** the sample. (d) If $P = 0.2$ then $n = 3.8416 \times 0.16/0.0025 = 245.9 \Rightarrow 246$, against 385 — about 139 extra interviews, 56% more than needed. Wasteful but *safely* so; use a pilot if one is available.

14.7

[8 marks]

(a) A relative margin r means an absolute margin $e = r\bar{Y}$, so

$$n_{WR} = \frac{z_{\alpha/2}^2 \sigma^2}{r^2 \bar{Y}^2} = \frac{z_{\alpha/2}^2}{r^2} \left(\frac{\sigma}{\bar{Y}} \right)^2 = \frac{z_{\alpha/2}^2 \text{CV}_Y^2}{r^2},$$

where $\text{CV}_Y = \sigma/\bar{Y}$ is the CV of the population. Only the ratio matters.

(b) (i) **Scale invariance** — the CV is unit-free, unchanged by inflation, currency, or measuring in thousands, so an old σ is stale while the old CV usually is not. (ii) **Empirical stability** — for economic variables the spread grows roughly proportionally with the level, so the CV stays nearly constant over time and across regions.

14.8

[12 marks]

(a) $t = \frac{\bar{y} - \mu_0}{\text{SE}(\bar{y})}$, reject if $t > z_\alpha$. This uses the Hájek normal approximation and (for now) pretends SE is known.

(b) Under H_1 , $\bar{y}$ has mean μ_1 :

$$1 - \beta = \mathbb{P}\left(Z > z_\alpha - \frac{\Delta}{\text{SE}}\right) \implies z_\alpha - \frac{\Delta}{\text{SE}} = -z_\beta \implies \text{SE} = \frac{\Delta}{z_\alpha + z_\beta}.$$

With $\text{SE} = \sigma/\sqrt{n}$, $n_{WR} = \frac{(z_\alpha + z_\beta)^2 \sigma^2}{\Delta^2}$. ■

(c) Two-sided: replace z_α by $z_{\alpha/2}$ (approximate, neglecting the wrong-tail probability). SRSWOR: $n_{WOR} \approx \frac{n_{WR}}{1 + n_{WR}/N}$ — the same correction as in the estimation approach.

(d) Δ is the **minimum detectable effect**: the smallest departure you care about. It is a subject-matter decision (a Rs. 500 jump in rent; a Rs. 50 rise in fuel cost that wipes out one delivery's payout). Since $n \propto 1/\Delta^2$, any non-zero difference is detectable with a big enough sample, so without a substantively chosen Δ the question “what n ?” has no answer.

14.9

[10 marks]

Solution. $\Delta = 5$, $\sigma = 9.8$, $z_{\alpha/2} = 1.96$, $z_\beta = 0.8416$. **Assumptions:** SRSWR (no fpc), Hájek approximation, σ known at the historical value, simple alternative $\bar{Y} = 100$.

$$n_{WR} = \frac{(1.96 + 0.8416)^2 (9.8)^2}{5^2} = \frac{(2.8016)^2 \times 96.04}{25} = \frac{7.849 \times 96.04}{25} = \frac{753.8}{25} = 30.15.$$

Round up: $\boxed{n = 31}$. (With a finite $N = 200$ this would fall to $31/(1 + 31/200) \approx 27$.)

14.10

[10 marks]

Solution. $P_0 = 0.26$, $\Delta = 0.05$, $z_{\alpha/2} = 1.96$, $z_\beta = 1.2816$. Under H_0 the variance is known: $P_0(1 - P_0) = 0.1924$ — **nothing has to be guessed**, which is the attraction of the proportion case.

$$n_{WR} = \frac{(1.96 + 1.2816)^2 \times 0.1924}{0.05^2} = \frac{10.508 \times 0.1924}{0.0025} = \frac{2.0218}{0.0025} = 808.75 \implies \boxed{n = 809}.$$

Detecting a 5 percentage-point shift needs a sample in the high hundreds — proportions are noisy, which is why epidemiological studies are large.

14.11

[8 marks]

Solution. Since $n \propto \frac{(z_\alpha + z_\beta)^2 \sigma^2}{\Delta^2}$ there are three levers:

Lever	Effect	Cost
Increase α	$z_\alpha \downarrow$, $n \downarrow$	more <i>false positives</i> ; rarely acceptable
Accept lower power	$z_\beta \downarrow$, $n \downarrow$	more <i>false negatives</i> ; an underpowered study may be a waste
Increase Δ	$n \propto 1/\Delta^2$ strongest lever	you can now detect only larger changes

Choose Δ : it is the only lever that is an honest scientific statement (“we care only about changes of at least this size”) rather than a degradation of the inference, and it has quadratic leverage. Raising α is worst, since false positives propagate into published claims. If none of the three suffices, the correct conclusion is the one the notes state bluntly: *consider whether the survey should be conducted at all*.

15 Solutions: stratified random sampling

15.1

[8 marks]

(a) $\bar{y}$ is not biased. $\mathbb{E}(\bar{y}) = \bar{Y}$ holds under SRSWOR whatever the sample happens to look like. What has gone wrong is *variance*, not bias.

“Representative **in expectation**” means that *averaged over repeated sampling* the composition matches the population — here $\mathbb{E}(\# \text{ men}) = 5$. “Representative **in this sample**” is a different claim, and SRS does not make it. With $n = 10$ from a 50/50 population, $\mathbb{P}(9 \text{ or } 10 \text{ of one sex}) \approx 2 \times \frac{11}{1024} \approx 2\%$: about one sample in fifty is this lopsided.

(b) **Stratify by gender**: treat men and women as two strata and draw an independent SRSWOR of 5 from each. The gender split is then fixed at the population value by construction, not left to chance. This requires **auxiliary information** — the gender of every unit in the frame — which SRS is completely blind to.

(c) **Yes**. Every unit still has a known, strictly positive inclusion probability ($\pi_{hj} = n_h/N_h = 5/500 = 1/100$ here, in fact equal across strata), so design-based inference remains available. Stratification restricts *which* samples are possible; it does not abandon randomisation.

15.2

[12 marks]

(a) $\bar{y} = \frac{1}{n} \sum_h \sum_i y_{hi}$ has

$$\mathbb{E}(\bar{y}) = \frac{1}{n} \sum_h n_h \bar{Y}_h = \sum_h \frac{n_h}{n} \bar{Y}_h.$$

This equals $\bar{Y} = \sum_h W_h \bar{Y}_h$ for *all* populations iff $\frac{n_h}{n} = W_h = \frac{N_h}{N}$ for every h — that is, iff the allocation is **proportional**. (It is also unbiased in the degenerate case $\bar{Y}_1 = \dots = \bar{Y}_H$.) In general it is **biased**, because the sample over-represents strata that were allocated more than their share.

(b) $\mathbb{E}(\bar{y}_{st}) = \frac{1}{N} \sum_h N_h \mathbb{E}(\bar{y}_h) = \frac{1}{N} \sum_h N_h \bar{Y}_h = \bar{Y}$ ■ — the weights N_h/N are exactly what undoes the unequal allocation.

(c) The SRSWORs are drawn **independently** across strata, so the $\bar{y}_h$ are independent and all cross-covariances vanish:

$$\text{Var}(\bar{y}_{st}) = \frac{1}{N^2} \sum_h N_h^2 \text{Var}(\bar{y}_h) = \frac{1}{N^2} \sum_h N_h^2 (1 - f_h) \frac{S_h^2}{n_h}, \quad f_h = \frac{n_h}{N_h},$$

using the within-stratum SRSWOR variance from Question 4.4. **Independence is used exactly once**, in dropping the covariances — say so.

(d) Since $\mathbb{E}(s_h^2) = S_h^2$ within each stratum,

$$\widehat{\text{Var}}(\bar{y}_{st}) = \frac{1}{N^2} \sum_h N_h^2 (1 - f_h) \frac{s_h^2}{n_h}$$

is unbiased, and the approximate CI is $\bar{y}_{st} \pm z_{\alpha/2} \sqrt{\widehat{\text{Var}}(\bar{y}_{st})}$.

15.3

[10 marks]

(a) **Equal** $n_h = n/H$: ignores everything; sensible only if the strata are similar in size and spread. **Proportional** $n_h = nW_h$: accounts for stratum *sizes*; the most frequently used. **Neyman** $n_h \propto N_h S_h$: accounts for sizes *and* within-stratum variability — sample more where there is more to learn.

(b) $f_h = \frac{n_h}{N_h} = \frac{nW_h}{N_h} = \frac{n}{N} = f$ for every h , so

$$\text{Var}_{prop} = \frac{1-f}{N^2} \sum_h \frac{N_h^2 S_h^2}{nW_h} = \frac{1-f}{N^2 n} \sum_h N N_h S_h^2 = \frac{1-f}{n} \sum_h W_h S_h^2.$$

(c) With $n_h = \frac{nN_hS_h}{\sum_{h'} N_{h'}S_{h'}}$,

$$\sum_h \frac{N_h^2 S_h^2}{n_h} = \frac{\sum_{h'} N_{h'} S_{h'}}{n} \sum_h N_h S_h = \frac{(\sum_h N_h S_h)^2}{n},$$

so, since $\text{Var}(\bar{y}_{st}) = \frac{1}{N^2} [\sum_h \frac{N_h^2 S_h^2}{n_h} - \sum_h N_h S_h^2]$,

$$\text{Var}_{Ney} = \frac{1}{n} \left(\sum_h W_h S_h \right)^2 - \frac{1}{N} \sum_h W_h S_h^2.$$

(d) Subtracting,

$$\text{Var}_{prop} - \text{Var}_{Ney} = \frac{1}{n} \left[\sum_h W_h S_h^2 - \left(\sum_h W_h S_h \right)^2 \right] = \frac{1}{n} \text{Var}_W(S_h) \geq 0,$$

the W -weighted variance of the numbers S_h (non-negative by Jensen or Cauchy–Schwarz). **Equality iff all S_h are equal** — when the strata are equally variable there is nothing for Neyman to exploit and it coincides with proportional allocation. ■

15.4

[10 marks]

Solution. Minimise $\text{Var}(\bar{y}_{st}) = \frac{1}{N^2} [\sum_h \frac{N_h^2 S_h^2}{n_h} - \sum_h N_h S_h^2]$ over $n_1, \dots, n_H$ subject to $\sum_h n_h = n$. The second sum is constant, so it suffices to minimise $g = \sum_h \frac{N_h^2 S_h^2}{n_h}$.

Lagrange. With $\mathcal{L} = \sum_h \frac{N_h^2 S_h^2}{n_h} + \lambda(\sum_h n_h - n)$,

$$\frac{\partial \mathcal{L}}{\partial n_h} = -\frac{N_h^2 S_h^2}{n_h^2} + \lambda = 0 \implies n_h = \frac{N_h S_h}{\sqrt{\lambda}} \propto N_h S_h,$$

and imposing $\sum_h n_h = n$ fixes the constant: $n_h = \frac{nN_h S_h}{\sum_{h'} N_{h'} S_{h'}}$. ■

Cauchy–Schwarz (cleaner, and gives the bound directly).

$$\left(\sum_h N_h S_h \right)^2 = \left(\sum_h \frac{N_h S_h}{\sqrt{n_h}} \cdot \sqrt{n_h} \right)^2 \leq \left(\sum_h \frac{N_h^2 S_h^2}{n_h} \right) \left(\sum_h n_h \right) = n \sum_h \frac{N_h^2 S_h^2}{n_h},$$

so $g \geq \frac{(\sum_h N_h S_h)^2}{n}$ always, with equality iff $\frac{N_h S_h}{\sqrt{n_h}} \propto \sqrt{n_h}$, i.e. $n_h \propto N_h S_h$. ■

15.5

[10 marks]

(a) Minimise $\text{Var}(\bar{y}_{st})$ subject to $A + \sum_h n_h C_h \leq C$. At the optimum the budget binds, so the constraint is $\sum_h n_h C_h = C - A$.

(b) With $\mathcal{L} = \sum_h \frac{N_h^2 S_h^2}{n_h} + \lambda \sum_h n_h C_h$,

$$-\frac{N_h^2 S_h^2}{n_h^2} + \lambda C_h = 0 \implies n_h = \frac{N_h S_h}{\sqrt{\lambda C_h}} \propto \frac{N_h S_h}{\sqrt{C_h}}.$$

Writing $n_h = t \frac{N_h S_h}{\sqrt{C_h}}$ and imposing $\sum_h n_h C_h = C - A$ gives $t \sum_h N_h S_h \sqrt{C_h} = C - A$, hence

$$n_h = (C - A) \cdot \frac{N_h S_h / \sqrt{C_h}}{\sum_{h'} N_{h'} S_{h'} \sqrt{C_{h'}}}.$$

Trap — a slip in the lecture slide

The Class-7 slide writes the denominator as $\sum_{h'} N_{h'} S_{h'} / \sqrt{C_{h'}}$. With that denominator the formula gives $\sum_h n_h = C - A$ (a sample size equal to an amount of money), not $\sum_h n_h C_h = C - A$. The correct normaliser is $\sum_{h'} N_{h'} S_{h'} \sqrt{C_{h'}}$, as derived above. Worth checking with the professor; the *proportionality* $n_h \propto N_h S_h / \sqrt{C_h}$ is not in doubt.

(c) If $C_h \equiv C_0$ then $n_h = (C - A) \frac{N_h S_h / \sqrt{C_0}}{\sqrt{C_0} \sum_{h'} N_{h'} S_{h'}} = \frac{(C-A)/C_0 \cdot N_h S_h}{\sum_{h'} N_{h'} S_{h'}}$, and since the affordable total is $n = (C - A)/C_0$, this is exactly $n_h = \frac{n N_h S_h}{\sum_{h'} N_{h'} S_{h'}}$ — **Neyman**. ✓

Intuition: n_h trades information against price. $N_h S_h$ measures how much variance reduction a unit from stratum h buys; $\sqrt{C_h}$ discounts it by what that unit costs. An expensive stratum must therefore be sampled less, and the discount is a *square root* because variance falls like $1/n_h$ while cost grows like n_h .

15.6**[12 marks]**

(a) The first term $\sum_h W_h (\bar{Y}_h - \bar{Y})^2$ is the **between-stratum sum of squares** (BSS): how far the stratum means are from the overall mean. The second, $\frac{1}{N} \sum_h (1 - W_h) S_h^2$, is a **within-stratum** (WSS) correction of order $1/N$.

(b) Stratification helps ($\text{Var}_{\text{SRS}} > \text{Var}_{\text{st}}$) exactly when

$$\sum_h W_h (\bar{Y}_h - \bar{Y})^2 > \frac{1}{N} \sum_h (1 - W_h) S_h^2,$$

i.e. when the **between-stratum variation is large relative to the within-stratum term**. In words: the strata must genuinely differ from each other in the study variable.

(c) If $\bar{Y}_1 = \dots = \bar{Y}_H$ the first term is 0, so the difference is *negative*: **SRSWOR is better**. Nothing is wrong with the theory — the **strata were formed badly**. They divide the population in a way that carries no information about Y , so stratification buys nothing while still costing the small within-term. **Moral: stratify on a variable correlated with the study variable.**

(d) In realistic surveys one stratifies on something known to be related to Y (region, sector, size class), so the BSS term is substantial, while the WSS term carries the factor $\frac{1}{N}$ and is negligible for large N . Hence stratification is “almost always beneficial” — but that is an empirical statement about sensible strata, not a theorem.

15.7**[12 marks]**

Solution. $W = (0.2, 0.3, 0.5)$, $f = 100/1000 = 0.1$.

(a) **Proportional:** $n_h = 0.1 N_h$, giving $\boxed{(20, 30, 50)}$.

Neyman: $N_h S_h = (12000, 12000, 10000)$ with sum 34000, so

$$n_h = 100 \cdot \frac{N_h S_h}{34000} = (35.29, 35.29, 29.41) \implies \boxed{(35, 35, 30)}.$$

Note the urban stratum gets 35 rather than 20: it is small but by far the most variable.

(b) $\sum W_h S_h^2 = 0.2(3600) + 0.3(1600) + 0.5(400) = 720 + 480 + 200 = 1400$; $\sum W_h S_h = 12 + 12 + 10 = 34$.

$$\text{Var}_{\text{prop}} = \frac{1-f}{n} \sum W_h S_h^2 = \frac{0.9}{100} (1400) = \boxed{12.6},$$

$$\text{Var}_{\text{Ney}} = \frac{34^2}{100} - \frac{1400}{1000} = 11.56 - 1.40 = \boxed{10.16}.$$

Check: $\text{Var}_{\text{prop}} - \text{Var}_{\text{Ney}} = \frac{1}{n} [1400 - 34^2] = \frac{244}{100} = 2.44$ ✓

(c) $\bar{Y} = 0.2(300) + 0.3(200) + 0.5(100) = 170$; $\sum W_h(\bar{Y}_h - \bar{Y})^2 = 0.2(130)^2 + 0.3(30)^2 + 0.5(70)^2 = 3380 + 270 + 2450 = 6100$, so $S^2 \approx 1400 + 6100 = 7500$ and

$$\text{Var}_{\text{SRSWOR}} = (1 - f) \frac{S^2}{n} = 0.9 \times \frac{7500}{100} = \boxed{67.5}.$$

Comparison: 67.5 (SRSWOR) $\gg$ 12.6 (proportional) $>$ 10.16 (Neyman). Stratification cuts the variance by a factor of more than 5 — almost all of it from the between-stratum term 6100, which stratification removes entirely. Neyman then shaves a further 19% off proportional by exploiting $S_1 \gg S_3$.

(d) Round to the nearest integers keeping $\sum n_h = n$ (here $35.29, 35.29, 29.41 \rightarrow 35, 35, 30$). Problems: rounding is not guaranteed optimal; it may violate $n_h \geq 1$ or $n_h \leq N_h$; and with several study variables the optimal allocations conflict. The principled fix is to solve the allocation as an **integer** programme — Wright's (2017) greedy algorithm, the R package **allocation**, or Brito et al. (2015) for multiple study variables.

15.8

[8 marks]

Solution. Beyond variance reduction:

1. **Strata-level parameters are wanted in their own right.** E.g. average income *within each district*, or infection rates separately for males and females. SRS gives no guarantee of enough units per subgroup.
2. **Administrative convenience and lower cost.** Fieldwork can be organised region-by-region, cutting travel and supervision costs.
3. **Different sampling frames per stratum.** A corporate database for big companies, street maps for local vendors — no single frame need exist.
4. **Different field procedures per stratum.** Email surveys for hospital staff, face-to-face interviews for rural patients.

And a good stratification *protects against a badly unrepresentative sample*, which is the motivation the course opens with.

16 Solutions: systematic sampling

16.1

[10 marks]

(a) With $N = nk$: draw i uniformly from $\{1, 2, \dots, k\}$ and take $\{i, i+k, i+2k, \dots, i+(n-1)k\}$. k is the **sampling interval**.

(b) There are exactly k possible samples $\mathbf{s}_1, \dots, \mathbf{s}_k$, so $\mathcal{S} = \{\mathbf{s}_1, \dots, \mathbf{s}_k\}$ with $p(\mathbf{s}_i) = \frac{1}{k}$. Every unit belongs to exactly one of them (the one indexed by its residue mod k), so

$$\pi_j = \frac{1}{k} = \frac{n}{N} \quad \text{for every } j.$$

(c) Two units lie in the same sample iff they are congruent mod k . Hence

$$\pi_{jj'} = \begin{cases} \frac{1}{k} = \frac{n}{N}, & j \equiv j' \pmod{k}, \\ 0, & \text{otherwise.} \end{cases}$$

Some second-order inclusion probabilities are zero, and this single fact is fatal for variance estimation: an unbiased design-based variance estimator (Horvitz–Thompson) requires $\pi_{jj'} > 0$ for every pair, since pairs that can never appear together carry no information. Concretely, you only ever observe one cluster, so nothing in the data speaks to the variation *between* clusters — which is exactly what $\text{Var}(\bar{y})$ is.

16.2

[12 marks]

(a) The k possible samples partition U into k disjoint groups of size n — call them clusters. Choosing i uniformly is choosing **one cluster at random**, and every unit of that cluster is then measured. So systematic sampling is cluster sampling with *one* cluster selected and complete enumeration inside it.

(b) With $\bar{Y}_i = \frac{1}{n} \sum_j Y_{ij}$ the i -th cluster mean, $\bar{y} = \bar{Y}_I$ where I is uniform on $\{1, \dots, k\}$, so

$$\mathbb{E}(\bar{y}) = \frac{1}{k} \sum_{i=1}^k \bar{Y}_i = \frac{1}{nk} \sum_i \sum_j Y_{ij} = \bar{Y}. \quad \square$$

(c) Since $\bar{y}$ takes the value $\bar{Y}_i$ with probability $\frac{1}{k}$,

$$\text{Var}(\bar{y}) = \frac{1}{k} \sum_{i=1}^k (\bar{Y}_i - \bar{Y})^2 = \sigma_b^2,$$

the **between-cluster** variance. ■

(d) $\text{Var}(\bar{y})$ depends on *nothing but* how the cluster means differ. To make it small we want the clusters to **resemble one another**; since the total variability of Y is fixed, that forces each cluster to be internally **heterogeneous** — each systematic sample must span the whole spread of the population.

This is the exact opposite of stratified sampling, where we want strata internally homogeneous and far apart from each other. Worth one sentence: in stratification we *sample from* every group, so we want the groups tight; in systematic sampling we *take one group entire*, so we want that group to look like the whole population.

16.3

[10 marks]

(a) $\text{Var}_{sys} = \frac{\sigma^2}{n} \{1 + (n-1)\rho_c\}$ and $\text{Var}_{WR} = \frac{\sigma^2}{n}$, so

$$\text{Var}_{sys} \leq \text{Var}_{WR} \iff (n-1)\rho_c \leq 0 \iff \rho_c \leq 0 \quad (n > 1).$$

(b) Using $\text{Var}_{sys} = \frac{S^2}{n} \cdot \frac{N-1}{N} [1 + (n-1)\rho_c]$ and $\text{Var}_{WOR} = \frac{S^2}{n} \cdot \frac{N-n}{N}$,

$$\text{Var}_{sys} \leq \text{Var}_{WOR} \iff (N-1)[1 + (n-1)\rho_c] \leq N-n \iff (n-1)\rho_c \leq \frac{N-n}{N-1} - 1 = \frac{1-n}{N-1},$$

and dividing by $n-1 > 0$ gives $\rho_c \leq -\frac{1}{N-1}$. ■

(c) $\text{Var}(\bar{y}) \geq 0$ forces $1 + (n-1)\rho_c \geq 0$, i.e. $\rho_c \geq -\frac{1}{n-1}$; and $\rho_c \leq 1$ as a correlation.

- $\rho_c = 1$: all units within a cluster are identical — maximum internal homogeneity. Then $\text{Var}(\bar{y}) = \sigma^2$, i.e. n times the SRSWR variance. **The worst case.**
- $\rho_c = -\frac{1}{n-1}$: then $\text{Var}(\bar{y}) = 0$ — every cluster has exactly the mean $\bar{Y}$, so the systematic sample is *always* right. **The ideal.**
- Note ρ_c cannot reach -1 unless $n = 2$.

16.4

[12 marks]

Solution. $N = 12$, $n = 3$, $k = 4$; $\bar{Y}$, σ^2 , S^2 as computed below.

(a) **Population A**, $Y = (1, 2, \dots, 12)$. Clusters (by starting point):

Start i	Units	Values, mean
1	1, 5, 9	1, 5, 9; $\bar{Y}_1 = 5$
2	2, 6, 10	2, 6, 10; $\bar{Y}_2 = 6$
3	3, 7, 11	3, 7, 11; $\bar{Y}_3 = 7$
4	4, 8, 12	4, 8, 12; $\bar{Y}_4 = 8$

$\bar{Y} = 6.5$, so

$$\text{Var}_{sys} = \frac{1}{4}[(1.5)^2 + (0.5)^2 + (0.5)^2 + (1.5)^2] = \frac{1}{4}(5) = \boxed{1.25}.$$

Here $\sum(Y_j - \bar{Y})^2 = 143$, so $\sigma^2 = 143/12 = 11.917$ and $S^2 = 143/11 = 13$:

$$\text{Var}_{WR} = \frac{\sigma^2}{n} = 3.972, \quad \text{Var}_{WOR} = (1 - \frac{3}{12})\frac{13}{3} = 3.25.$$

Systematic sampling is **far better** than both.

(b) **Population B**, $Y = (10, 20, 30, 40)$ repeated three times. Now cluster i consists entirely of one repeated value: $\bar{Y}_i = 10, 20, 30, 40$. With $\bar{Y} = 25$,

$$\text{Var}_{sys} = \frac{1}{4}[15^2 + 5^2 + 5^2 + 15^2] = \frac{500}{4} = \boxed{125}.$$

Here $\sigma^2 = 125$ and $S^2 = 1500/11 = 136.36$, so $\text{Var}_{WR} = 41.67$ and $\text{Var}_{WOR} = 34.09$. Systematic sampling is **catastrophically worse**.

(c) From $\text{Var}_{sys} = \frac{\sigma^2}{n}\{1 + (n-1)\rho_c\}$:

$$\text{A: } 1.25 = 3.972(1 + 2\rho_c) \Rightarrow \rho_c = -0.343; \quad \text{B: } 125 = 41.67(1 + 2\rho_c) \Rightarrow \rho_c = 1.$$

Explanation. In A the frame is *sorted*, so every systematic sample picks one small, one medium and one large value: clusters are internally heterogeneous and nearly identical to one another. Since $-0.343 \leq -\frac{1}{N-1} = -0.0909$, systematic beats SRSWOR — as observed.

In B the frame is *periodic with period exactly* $k = 4$, so each cluster contains a single repeated value: perfectly homogeneous internally (ρ_c attains its maximum 1) and maximally different from the others. This is the classic systematic-sampling disaster.

16.5

[10 marks]

(a) A systematic sample is **one cluster**. At the cluster level the sample size is 1, and no variance can be estimated from a single observation. Formally $\text{Var}(\bar{y}) = \sigma_b^2$ involves all k cluster means and you observe exactly one; equivalently, $\pi_{jj'} = 0$ for units in different clusters (Question 8.1(c)), so no design-unbiased estimator exists.

(b) Comparing

$$\mathbb{E}(\hat{V}) = (1-f)\frac{S^2}{n} \cdot \frac{N-1}{N}(1-\rho_c) \quad \text{with} \quad \text{Var}(\bar{y}) = \frac{S^2}{n} \cdot \frac{N-1}{N}[1 + (n-1)\rho_c],$$

and writing $c = \frac{S^2}{n} \frac{N-1}{N} > 0$,

$$\mathbb{E}(\hat{V}) - \text{Var}(\bar{y}) = c[(1-f)(1-\rho_c) - 1 - (n-1)\rho_c] = c[-f - \rho_c(n-f)].$$

The bracket vanishes when $\rho_c = -\frac{f}{n-f} = -\frac{1}{N-1}$ (using $f = n/N$), confirming the stated unbiasedness condition. Moreover, since $n-f > 0$:

$$\rho_c < -\frac{1}{N-1} \Rightarrow \hat{V} \text{ over-estimates}; \quad \rho_c > -\frac{1}{N-1} \Rightarrow \hat{V} \text{ under-estimates}.$$

(c) It is **conservative exactly when** $\rho_c < -\frac{1}{N-1}$ — which by Question 8.3(b) is precisely the region where systematic sampling is genuinely beating SRSWOR, i.e. when the clusters are heterogeneous enough. Then $\hat{V}$ overstates the uncertainty and the resulting CIs are too wide, which is the safe direction. The danger is the common case $\rho_c > -\frac{1}{N-1}$ (clusters more homogeneous than neutral): there $\hat{V}$ under-states the true variance and the CIs are falsely narrow.

16.6

[10 marks]

Solution. (i) **m independent systematic samples (IPNS, after Mahalanobis).** Draw $m > 1$ independent random starts, giving systematic samples of size n each with means $\bar{y}^{(1)}, \dots, \bar{y}^{(m)}$. Then

$$\bar{\bar{y}} = \frac{1}{m} \sum_{t=1}^m \bar{y}^{(t)} \text{ is unbiased for } \bar{Y}, \quad \hat{V} = \frac{1}{m(m-1)} \sum_{t=1}^m (\bar{y}^{(t)} - \bar{\bar{y}})^2$$

is an unbiased estimator of $\text{Var}(\bar{\bar{y}})$. **Cost:** you now measure mn units — you need m times the budget.

(ii) **m random starts within the first mk units.** Suppose $m \mid n$ and write $N = \frac{n}{m} \times mk$. Choose m independent starts from $\{1, \dots, mk\}$ and from each take every mk -th unit, giving m systematic samples of size n/m . Apply the same formulas. **Cost:** the total sample is still n , but each subsample is smaller, so $\hat{V}$ is based on m noisier means and is itself imprecise; it also requires $m \mid n$.

(iii) **Wolter's (1984) successive-difference estimator.** Using only the one sample,

$$\hat{V} = \frac{1-f}{n} \left[\frac{1}{2(n-1)} \sum_{j=1}^{n-1} (Y_{i,j+1} - Y_{ij})^2 \right].$$

Assumption: an autocorrelation structure along the list — essentially that the population varies smoothly with position, so neighbouring sampled units differ only by noise. No extra data is needed, which is its appeal; but if the frame is periodic the assumption fails badly.

(A cheaper fourth option: split the single systematic sample of size n into m disjoint subsamples of size n/m and pretend they are independent systematic samples. It costs nothing, but the pretence is simply false.)

16.7

[8 marks]

(a) $k = N/n = 14/4 = 3.5$ is not an integer, so “every k -th unit” is undefined.

- $k = 3$: starts 1, 2, 3 give $\{1, 4, 7, 10, 13\}$, $\{2, 5, 8, 11, 14\}$, $\{3, 6, 9, 12\}$ — sizes 5, 5, 4, mean size $14/3 = 4.67 \neq 4$.
- $k = 4$: starts 1, 2, 3, 4 give $\{1, 5, 9, 13\}$, $\{2, 6, 10, 14\}$, $\{3, 7, 11\}$, $\{4, 8, 12\}$ — sizes 4, 4, 3, 3, mean size $3.5 \neq 4$.

In both cases π_j is still constant, but there is **no control over the sample size**, which is now random.

(b) **Circular systematic sampling.** Arrange the N units on a circle; pick a random start i from $\{1, \dots, N\}$ (not $\{1, \dots, k\}$); take k as the nearest integer to N/n ; then select, for $j = 0, 1, \dots, n-1$, the unit $i + jk$ if $i + jk \leq N$, and $i + jk - N$ otherwise.

With $N = 14$, $n = 4$, $k = 4$, start $i = 13$:

$$13, \quad 17 - 14 = 3, \quad 21 - 14 = 7, \quad 25 - 14 = 11 \quad \implies \quad \{13, 3, 7, 11\},$$

of size exactly 4.

(c) For each unit j there are exactly n starting points that capture it (namely $j, j-k, j-2k, \dots$ reduced mod N), and each start has probability $1/N$, so

$$\pi_j = \frac{n}{N} \quad \text{for every } j,$$

and the sample size is **always exactly** n . Circular systematic sampling therefore fixes both defects of the ad hoc fixes at once: equal inclusion probabilities *and* a fixed sample size, at the cost of N possible samples instead of k .

16.8

[8 marks]

Solution. Preferred when:

1. The population is a **flow rather than a file** — e.g. commuters at the Howrah ferry ghat in a time window: there is no list to sample from, but you can take every k -th arrival.
2. **Operational simplicity and speed**; there is no “collision” issue. In forestry, auditing or agriculture it is far easier to instruct a worker to measure every 10th tree than to navigate to random GPS coordinates.
3. The frame is **sorted by the study variable**, so the sample sweeps the whole spectrum — e.g. every 10th student from an admission list ordered by entrance score.
4. **Large-scale practice**: the 1940 US Census supplementary questions, CMFRI’s marine fish catch estimation, and geological, household and industrial surveys.

Goes badly wrong when:

1. The frame is **periodic with period related to k** . If males and females alternate and k is even, the sample is all one gender. Estimating average daily road traffic by sampling every 7th day always hits the same weekday.
2. There is **grouped segregation**: inspecting every 50th tree in a forest where a rare disease spreads in tight groups of 5–6 trees — the sample either misses the clusters entirely or, by bad luck, lands on them repeatedly.

17 Solutions: applied and mixed problems

17.1

[12 marks]

Solution. $N = 1200$, $n = 150$, $p = 42/150 = 0.28$, $f = 0.125$.

(a) $\widehat{\text{Var}}(p) = (1 - f) \frac{p(1-p)}{n-1} = 0.875 \times \frac{0.2016}{149} = 0.0011839$, so $\widehat{\text{SE}}(p) = 0.0344$.

(b) $0.28 \pm 1.96(0.0344) = 0.28 \pm 0.0674$, i.e. $[0.213, 0.347]$.

(c) $\hat{T} = 1200(0.28) = 336$ businesses, $\widehat{\text{SE}}(\hat{T}) = 1200(0.0344) = 41.3$.

(d) $\widehat{\text{CV}} = 0.0344/0.28 = 12.3\% \leq 16.6\%$: **reliable**.

(e) $e = 0.03$, $z^2 = 3.8416$. (i) *Worst case* $P = 0.5$: $n_{WR} = \frac{3.8416(0.25)}{0.0009} = 1067.1$, so $n_{WOR} = \frac{1067.1}{1+1067.1/1200} = 565$. (ii) *Pilot* $P = 0.28$: $n_{WR} = \frac{3.8416(0.2016)}{0.0009} = 860.5$, so $n_{WOR} = \frac{860.5}{1+860.5/1200} = 502$. The pilot saves about 63 interviews. Note how large the fpc effect is: with $N = 1200$, n_{WOR} is barely half of n_{WR} .

17.2

[10 marks]

Solution. The design is unusable.

Sampling errors / design failures (fatal):

1. **No frame, no randomisation** — this is convenience sampling. Selection probabilities are unknown, so no unbiased estimator, standard error or CI exists; the reported $s/\sqrt{120}$ is meaningless.
2. **Sampled $\neq$ target population**. Households spending *nothing* on tuition — exactly those that pull the average down — have **zero** chance of selection. The estimate is biased upward by construction and no n fixes it.
3. **Coverage bias**. “Well-known” centres are likely the expensive ones, biasing further upward.
4. **Unequal selection probabilities within the frame**. A household with several children at the centres has several chances of selection — an unequal-probability design being analysed as an equal-probability one.

Non-sampling errors:

5. **Non-response bias:** 40% refused, and willingness to discuss spending plausibly depends on how much is spent — the *Literary Digest* mechanism again.
6. **Measurement error:** the question leaves the reference period unspecified (month? year?), the unit ambiguous (one child or all?), and is a sensitive financial question inviting misreporting.

What should have been done: obtain a household frame (electoral roll, municipal list); draw an SRSWOR or a stratified sample; fix the margin of error in advance and solve for n ; pilot the questionnaire; plan callbacks to reduce non-response.

17.3

[10 marks]

Solution. $Y = (2, 4, 4, 6, 8, 12)$, $N = 6$.

(a) $\sum Y_j = 36$ so $\bar{Y} = 6$; deviations $-4, -2, -2, 0, 2, 6$ with squares summing to 64: $\sigma^2 = \frac{64}{6} = 10.667$, $S^2 = \frac{64}{5} = 12.8$. Check $\frac{6}{5}(10.667) = 12.8$ ✓

(b) The $\binom{6}{2} = 15$ sample means are

$\{1, 2\}:3$	$\{1, 3\}:3$	$\{1, 4\}:4$	$\{1, 5\}:5$	$\{1, 6\}:7$
$\{2, 3\}:4$	$\{2, 4\}:5$	$\{2, 5\}:6$	$\{2, 6\}:8$	$\{3, 4\}:5$
$\{3, 5\}:6$	$\{3, 6\}:8$	$\{4, 5\}:7$	$\{4, 6\}:9$	$\{5, 6\}:10$

They sum to 90, so $\mathbb{E}(\bar{y}) = 90/15 = 6 = \bar{Y}$ ✓. Their squares sum to 604, so $\mathbb{E}(\bar{y}^2) = 40.267$ and $\text{Var}(\bar{y}) = 40.267 - 36 = 4.267$. Formula: $(1 - f)\frac{S^2}{n} = \frac{2}{3} \cdot \frac{12.8}{2} = 4.267$ ✓

(c) $\text{Var}_{WR} = \frac{10.667}{2} = 5.333 > 4.267$ ✓, and the ratio $4.267/5.333 = 0.8 = \frac{N-n}{N-1} = \frac{4}{5}$ exactly, as Question 4.6(a) predicts.

17.4

[12 marks]

Solution. (a) **Choose stratified sampling by ward.**

- **SRSWOR** is dominated: you *know* the ward of every household, and SRS is blind to that auxiliary information. Its variance carries the full between-ward variability, which by hypothesis is the dominant component.
- **Stratified** by ward removes the between-ward term entirely — exactly the situation of Question 7.6(b), where BSS is large and WSS small, so stratification helps most.
- **Systematic** is attractive operationally and, on a ward-sorted list, behaves similarly (see (c)) — but its variance cannot be estimated unbiasedly, which matters for a survey that must report standard errors.

(b) **Proportional allocation** if the S_h are unknown: it needs only the N_h , which you have, and it guarantees $f_h = f$ and a self-weighting sample. **Neyman** if a pilot or previous study gives the S_h : it is at least as good and strictly better unless all S_h are equal. Neyman's extra requirement is precisely knowledge of the within-ward spreads (or at least their relative values).

(c) Every 50th household from a ward-sorted list takes roughly $N_h/50$ households from ward h — i.e. approximately **proportional allocation**. This is called *implicit stratification*, and here it is **good**: it captures most of the benefit of stratifying by ward for free. The catch is the one above: no unbiased variance estimator, so you would have to fall back on IPNS or on pretending it is SRSWOR (which, as Question 8.5 shows, may understate the variance).

(d) **Non-response** is the obvious exposure — households not at home, or refusing to disclose consumption — and it is plausibly related to consumption itself (high users may be reluctant). Mitigate with repeat callbacks at different times of day, and, better, by taking the figure from **meter readings** rather than recall, which removes the measurement-error component at the same time.

Constructed from Classes 1–9 (107 slides). No previous-year papers exist for this course; mark weights follow the stated 8–12 and 5 mark pattern.